

## UCE852: INTERNET OF THINGS (IoT) & SMART CITIES

| L | T | P | Cr  |
|---|---|---|-----|
| 3 | 1 | 0 | 3.5 |

**Course Objective:** This subject aims to develop an understanding of the application of Internet of things (IoT) to tame the pressure of urbanization, create new experience for city residents, make day-to-day living more comfortable and secure and to present an optimal approach to the implementation of smart city solutions.

### **Introduction:**

The evolution of cities up to the present day: social, political and spatial planning models, Urbanization and its impacts on cities, Urban evolution in India, Changing patterns of urban growth, Quality of life in the city, Efficiencies and inefficiencies in cities; challenges and opportunities, Eco challenges in the contemporary cities, Principles of green and smart cities, International initiatives including UN and EU level, Corporate social and environmental strategies in cities.

### **Smart Cities:**

Definition, drivers, barriers and benefits of smart cities, Characteristics and factors of Smart cities, Examples of smart cities, Study of the existing cities, Finding problems and how far they are solvable, Designing for Smart cities, Design, development and exhibition of a feasible innovation project which will enrich citizens and the city through all its phases: determining the scope, defining the idea, establishing objectives, identifying partners, selecting and acquiring tools and knowledge, planning and presentation, beginning to put the project into practice.

### **Internet of Things:**

IoT concepts: Technologies that led to evolution of IoT, IoT and SCADA, IoT and M2M, IoT and Big Data; IoT Standards: Requirement of international standard (case study), IoT standards in practice, Operating platforms /systems; Components of IoT System: Design of IoT systems; Development of prototypes.

### **IoT implementation in Smart Cities:**

IoT Applications: Lighting as a service (case study), Intelligent Traffic systems ( case study), Smart Parking ( case study), Smart water management ( case study) ; IoT for smart cities ( Case study Smart city Barcelona); IoT in Indian Scenario: IoT and Aadhaar, IoT for health services, IoT for financial inclusion ; IoT for rural empowerment.

### **IoT applications for Value Creations:**

IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications; Four Aspects in your Business to Master; Value Creation from Big Data and Serialization; IoT for Retailing Industry; IoT For Oil and Gas Industry; Opinions on IoT Application and Value for Industry, Home Management, e-Health.

### **Course Learning Outcomes (CLO):**

After the completion of this course the students would be able to:

1. Use the principles of green and smart cities, International initiatives including UN and EU level.
2. Determining the scope, defining the idea and establishing objectives for smart cities.
3. Develop the concept of Internet of Things and its implementation to smart cities.
4. Application of IoT for value creation in Industry.

**Text Books:**

1. *The Internet of Things: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World*
2. *Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014*
3. *Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013*
3. *CunoPfister, Getting Started with the Internet of Things.*

**Reference Books:**

1. *How Green is Cities? By Dimitri Devuyst, Colombia University Press.*
2. *New York Sustainability Science and Engineering Vol 1, By Martin A. Abraham (editor) Elsevier Publication,*

**Evaluation Scheme:**

| Sr. No. | Evaluation Elements                         | Weightage (%) |
|---------|---------------------------------------------|---------------|
| 1       | MST                                         | 30            |
| 2       | EST                                         | 45            |
| 3       | Sessional (May include assignments/quizzes) | 25            |